EYENET VERTICAL SOLUTION

MARITIME CREW SAFETY & FLEET MANAGEMENT

Eyenet for Shipping

Staff safety, presence tracking, and crew management for maritime fleets – with automated notifications to captain, officers, and shipping company HQ.

TYPE	DATE	PRODUCTS	DEPLOYMENT
Vertical Solution Overview	14 July 2026	5 Applications + Matrix	On-Premises / VPN Fleet Mesh

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01 About Eyenet for Shipping

Ships are isolated, safety-critical environments operating far from shore-based management. Eyenet brings computer vision-powered crew presence, restricted area protection, and automated multi-level notifications to every vessel – then connects the fleet through a secure VPN mesh.

The maritime industry faces operational challenges that off-the-shelf security systems cannot address. A vessel at sea is self-contained: crew must be accounted for, watch stations must stay manned, restricted zones must be enforced 24/7, and all of this must work with zero internet dependency. Each vessel runs a complete EyeNet installation on a single mini PC – fully offline at sea, synchronising with headquarters when docked or via satellite.

02 The Maritime Operational Challenge

Offline

CORE OPERATION

Ships operate without internet for days or weeks. Presence tracking, alerting, and access control must run locally. Data syncs when connectivity resumes.

Safety

CRITICAL ENVIRONMENT

Know who is onboard and where. Enforce restricted areas. Support lone-worker monitoring and muster headcount without manual roll calls.

Compliance

CREW & FLEET MANAGEMENT

Bridge shore-side ERP crew modules with actual onboard presence. Detect mismatches and report to captain, officers, and HQ.

03 Eyenet Presence – Crew Attendance & Muster

Dual-factor crew verification at critical stations – facial recognition plus QR/NFC badge – with automated attendance reports to captain and HQ.

Presence assigns crew to watch schedules and zones, then tracks attendance in real time. Missed watch check-ins alert immediately – no delayed discovery of a missing person.

Maritime use cases

- **Watch changeover verification** – dual-factor check-in at bridge and ECR; alert if the scheduled officer does not appear.
- **Muster station headcount** – Area mode gives live accounted/missing status without manual roll calls.
- **ERP crew module sync** – import schedules; export verified attendance when connected.
- **Certification-aware alerts** – flag STCW gaps before a watch begins.
- **Automated reports** – daily, per-voyage, and per-shift distribution to captain and fleet managers.

Modes

Area | Gate

Integrations

ERP Crew Module | Matrix

Notify: Captain, Chief Engineer, Chief Mate, Fleet HR

Offline: Full local operation; queue and sync at port.

04 Eyenet Gate – Gangway & Entrance Control

Real-time onboard/ashore headcount at every gangway – crew, visitors, contractors, and port officials.

Gate monitors embarkation and disembarkation with identity verification and automated logging. In emergencies the captain has an accurate headcount; during port calls visitors are tracked separately from crew.

Maritime use cases

- **Gangway monitoring** – onboard vs ashore with timestamps; shore-leave status into ERP.
- **Visitor & contractor tracking** – alert if temporary personnel remain at departure.
- **Stowaway detection** – unrecognised faces trigger alerts with image and location.
- **Abandon-ship picture** – Gate + Presence shows last-seen and muster arrival status.
- **Port state readiness** – automated crew lists with embark/disembark timestamps.

Modes

Gate

Integrations

ERP Crew Module | Presence | Matrix

Notify: Captain, Security Officer, Chief Mate, Port Agent

Offline: Local event log; sync manifests when connected.

05 Eyenet Sentinel – Watch Station Vigilance

Detect when a watch station is unattended – bridge, ECR, or security post – and escalate before it becomes an incident.

Sentinel monitors designated watchkeepers and escalates if no activity is detected beyond a configurable threshold. Purpose: prevent unattended watch – a leading cause of maritime incidents.

Maritime use cases

- **Bridge OOW monitoring** – mobile first alert, then captain / chief mate escalation.
- **UMS duty rounds** – missed engine-room rounds alert the chief engineer.
- **Security post monitoring** – support ISPS watchkeeping compliance.
- **Per-station parameters** – thresholds, escalation chains, and active hours per post.
- **Fatigue patterns** – combine with Presence to flag MLC hour risk early.

Modes

Area

Integrations

Presence · Orchestrator

Notify: Watch Officer → Captain / Chief Engineer / Chief Mate

Offline: Fully self-contained onboard detection and alerting.

06 Eyenet Vradar – Patrol & Restricted Zones

Wearable camera eyewear for onboard security patrols. Instant face recognition against crew manifest and watchlists – on-premises, no cloud.

Vradar equips the security officer with wearable cameras that verify faces against the crew manifest and persons-of-interest lists while on patrol. All processing stays on the vessel.

Maritime use cases

- **Manifest verification on patrol** – silent verify; alert on unrecognised faces.
- **Restricted zone enforcement** – bridge, ECR, cargo control, and other virtual perimeters 24/7.
- **Stowaway search** – accelerate ISPS pre-departure compartment searches.
- **Port security coordination** – load POI watchlists before port calls.
- **Drone mode** – aerial coverage for large decks and cargo holds.

Modes

Goggles · Drone

Integrations

ERP Crew Manifest · Gate · Presence

Notify: Security Officer, Captain, Chief Mate, CSO at HQ

Offline: Local recognition; manifests refreshed when connected.

07 Eyenet Signage – Crew & Emergency Displays

Digital boards for crew decks, mess halls, and muster stations – with vision-triggered emergency content switching.

Signage turns vessel displays into managed information boards. In an emergency, detection or muster activation switches content to evacuation routes and safety instructions.

Maritime use cases

- **Mess hall displays** – schedules, weather, ETA, safety bulletins.
- **Emergency override** – automatic switch to evacuation and muster guidance.
- **Safety campaign playback** – PPE, confined space, and fire prevention messaging.
- **Visitor orientation** – gangway displays for temporary personnel during port calls.

Modes

Spot | Gate | Area

Integrations

Fire Detect | Orchestrator | Matrix

Notify: All crew via displays; Captain override; HQ content push

08 Eyenet Matrix – Fleet-Wide Reporting

Connect every vessel through a secure VPN mesh. Centralised dashboards and fleet-wide visibility from one Matrix installation at headquarters.

Degraded mode: at sea, data is queued locally. When VPN reconnects, queued data synchronises automatically – no data loss, no single point of failure for onboard safety.

Matrix turns vessel installations into a unified fleet system. It queries vessels in parallel over the VPN mesh and aggregates attendance, security events, and operational analytics for shore-side managers.

Fleet use cases

- Fleet-wide crew attendance compliance
- Centralised users, roles, and vessel groups
- Cross-vessel POI / watchlist distribution
- Operational analytics across vessels and routes
- Regulatory audit packs (STCW, ISPS, MLC)

Administrative capabilities

- Aggregated fleet reports
- Fleet group management
- Centralised RBAC
- Consolidated audit logs
- Fleet-wide watchlist push
- Signage / bulletin content push

09 Automatic Notification Architecture

Three tiers: onboard alerts (zero latency), ship-to-HQ sync, and bi-directional ERP crew module integration.

1 Onboard

Zero latency. Local network only.

- Watch absence → push + bridge alarm
- Restricted zone → eyewear + captain
- Muster headcount → captain / chief mate
- Unrecognised gangway face → security + captain
- Visitor still onboard at departure

2 Ship → HQ

VPN in port; satellite for critical alerts.

- Daily attendance → fleet HR
- Security events → CSO
- Watch exceptions → fleet ops
- Certification expiry → crewing
- Queued sync when VPN returns

3 ERP Crew Module

Bi-directional shipping ERP sync.

- ERP → manifest, schedules, certs
- Eyenet → attendance & embark logs
- Manifest mismatch alerts
- Certification gap before watch
- MLC rest-hour violation flags

Delivery channels

Captain Dashboard · Officer Mobile Push · Email to HQ · Audible / Visual Alarm · ERP Webhook · Matrix Dashboard

Critical safety alerts (MOB, fire, abandon ship) notify all configured channels simultaneously.

10 Industrial Safety Coverage

PPE compliance

Detect missing hard hat, vest, glasses, or boots in machinery spaces, holds, and deck zones – configurable per area.

Presence · Sentinel

Lone-worker protection

Monitor solitary work in ECR, pump room, holds, and tanks. No-movement thresholds trigger duty-officer check-in.

Sentinel · Presence

Incident documentation

Auto-capture timestamped evidence, personnel, and zone data; sync reports to Matrix when connected.

Matrix · Orchestrator

Confined space entry

Track entry/exit headcount; alert if exits do not match entries within the permit window.

Gate · Presence

Fire & smoke integration

Vision-based detection can trigger alarms, signage override, and notifications to captain and fire team.

Fire Detect · Signage

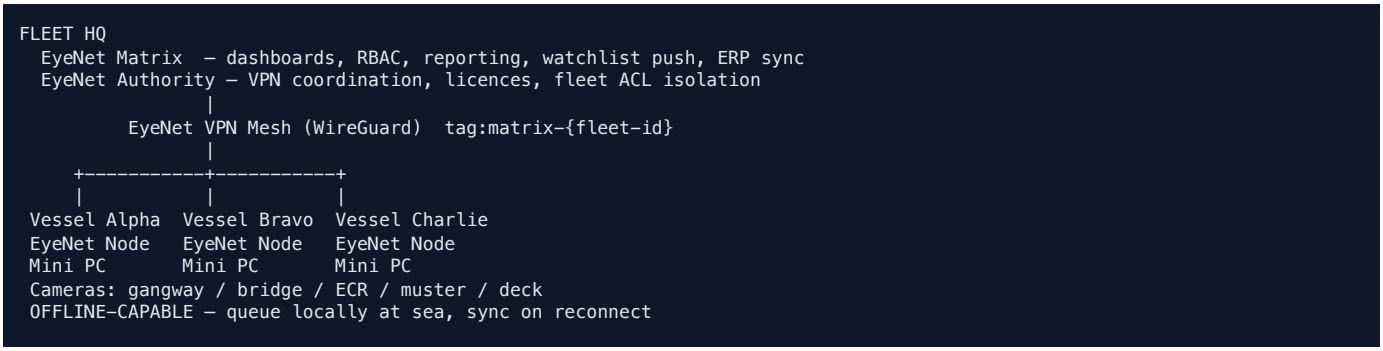
Muster & emergency response

Live movement toward muster, station headcount, and automated post-drill compliance reports.

Gate · Presence

11 VPN-Powered Fleet Connectivity

Each vessel runs a self-contained EyeNet node. The EyeNet VPN mesh connects vessels to HQ without exposing services to the public internet.



Architecture properties

- **Mesh topology** – vessels and HQ communicate directly; Matrix queries in parallel.
- **ACL fleet isolation** – fleet tags prevent cross-fleet communication on the shared coordination server.
- **Zero public exposure** – encrypted tunnels only; no public EyeNet service ports.
- **Automatic enrollment** – pre-auth key from Authority; discoverable by Matrix after join.
- **Degraded resilience** – offline vessels keep full local safety; Matrix marks them unreachable until sync.

Connectivity states

STATE	VESSEL	HQ
In port	Full VPN sync	Live visibility
Coastal	Priority alerts first	Critical live; rest next sync
Satellite	Bandwidth-optimised	Critical alerts
Offline	Full local ops	Unreachable; no data loss

12 Maritime Use Case Scenarios

SCENARIO	PRODUCTS	MODE	NOTIFY	ERP
Watch officer absent from bridge	Sentinel + Presence	Area	OOW → Captain → Chief Mate → Fleet Ops	Schedule in / attendance out
Missed muster in emergency	Gate + Presence	Gate · Area	Captain → Chief Mate → DPA	Manifest + muster report
Unauthorised person in engine room	Presence + Vradar	Area · Goggles	Chief Engineer → Security → Captain	Certification check
Stowaway in pre-departure search	Vradar + Gate	Goggles · Gate	Security → Captain → CSO → Port	Visitor / POI match
Watch hours exceed MLC limits	Presence + Sentinel	Area	Captain → Fleet HR → DPA	Hours vs schedule
Visitor not disembarked	Gate	Gate	Captain → Security → Port Agent	Visitor log / clearance
Smoke in cargo hold	Fire Detect + Signage + Orchestrator	Area · Spot	Captain → Fire Team → Crew → HQ	Incident + cargo cross-ref
Lone worker unresponsive	Sentinel	Area	Duty Officer → Captain → Medical	Permit-to-work check
Fleet attendance compliance review	Matrix + Presence	Area · Gate	Fleet HR → Fleet Ops Director	Aggregated export
Port state crew documentation	Gate + Presence + Matrix	Gate · Area	Captain (on-demand)	Crew list, hours, certs

13 Operating Modes Reference

MODE	DESCRIPTION	MARITIME APPLICATION
Spot	Fixed close-range monitoring	Mess displays, visitor kiosks
Gate	Entrances, exits, corridors	Gangway, confined space, muster approaches
Area	Wide-area rooms and outdoor spaces	Bridge, ECR, holds, muster, deck
Goggles	Wearable first-person monitoring	Patrols, stowaway searches, zone enforcement
Drone	Aerial mobile coverage	Large decks, cargo holds, tanker perimeter

14 Technical Specifications

1

MINI PC PER VESSEL

16–32 GB RAM, modern multi-core CPU, standard GPU. Fanless form factor; 12–24V DC or 110–240V AC.

2–5

CAMERAS PER VESSEL

Standard IP, WiFi, or USB cameras. Typical: gangway, bridge, ECR, muster, deck/cargo.

100%

OFFLINE CAPABLE

Core safety and presence functions run fully offline. Connectivity affects HQ visibility only.

All applications run on-premises. Video and presence data never leave the vessel unless explicitly configured. The VPN mesh provides encrypted fleet connectivity without public exposure.

15 Note

This document describes how the existing Eyenet product suite — Presence, Gate, Sentinel, Vradar, Signage, and Matrix — applies to maritime shipping. No new products or custom development are required. On-premises architecture, offline capability, VPN mesh connectivity, and orchestrator-driven notifications are inherently suited to maritime operations. Shipping ERP crew-module integration is supported through the existing API gateway and webhook infrastructure.

Prepared by EyeNet for informational purposes. Product specifications and availability subject to change.